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Intraclonal diversification and cell competition shape evolutionary trajectories in glioblastoma

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Tumor evolution is widely recognized as a major driver of resistance, but it may also represent a novel therapeutic opportunity if its underlying rules can be deciphered and hijacked. We previously showed that glioblastoma progression is characterized by extensive clonal extinction driven by cell competition, a process that continues within established clones and progressively reshapes their composition. Here, we investigated how phenotypic diversity arises from the progeny of a single glioma cell.

Cells derived from monoclonal murine gliomas were labeled with transcribed genetic barcodes and profiled by single-cell RNA sequencing at baseline. Subclonal dynamics were then followed by longitudinal barcode sequencing across multiple independent expansion experiments, allowing direct reconstruction of evolutionary trajectories and their relationship with transcriptional states.

Although all cells originated from a single founder, subclones diversified into distinct phenotypic states. Across independent evolution experiments, we observed the emergence of stable high-fitness populations, the replacement between competing states, and transient populations that expanded before progressively disappearing. These patterns were highly consistent across experiments, suggesting that cell-intrinsic determinants commit subclones toward reproducible long-term evolutionary trajectories.

Transcriptional profiling identified gene programs associated with the current fitness of individual subclones, but did not predict their future evolutionary behavior, despite the predetermination we observed. These findings suggest even if the determinants of future fitness are already established within tumor populations, they are not reflected in their transcriptional state.

Our results indicate that glioblastoma evolution is shaped by the continuous generation of phenotypic diversity followed by reproducible fitness-based selection. Rather than representing a purely stochastic process, tumor evolution appears to follow exploitable biological constraints.

Defining the mechanisms underlying these hidden fitness states may provide a framework to manipulate clonal competition and direct tumor evolution toward more therapeutically vulnerable conditions.



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